DRAWING COMPLIANCE CHECKER

A release gate for drawings: concept demo for GRSE GAINS 2026

Prepared 12 October 2026

Project Overview and Flow

What the project is, and how a drawing moves through it

> This document explains the project to someone who has not seen it: the problem, the idea, who uses the tool, and what happens to a drawing from upload to release. It needs no programming knowledge. The companion document, *Code Explanation*, covers how it is built.

> **Everything in this project is mock data.** The drawings, the drawing register, the bills of materials, the work orders and the rules with their clause numbers were invented for the demonstration. None of it comes from GRSE or from a classification society.

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1. The problem, and what the tool does about it

A shipyard cuts steel from drawings, and a drawing changes over time: revision A is replaced by revision B. A work order or a bill of materials is released to the shop floor against one particular revision. If that revision is out of date or was never approved, the wrong material is ordered or cut. The ERP *stores* which revision is current, but it does not *stop* anyone using the wrong one.

The Drawing Compliance Checker is a **release gate** that sits before that release. For each drawing it:

- reads the drawing's title block and compares its revision and status with the approved drawing register;
- runs a small set of fixed compliance checks, each tied to a numbered clause of a standard;
- shows exactly what changed since the previous revision;
- checks the parts list against the ERP's bill of materials, and lists the work orders and purchase orders a revision change affects;
- gives a verdict (**Pass**, **Needs review** or **Blocked**) with a reason for every finding;
- leaves the final decision to a named human reviewer, and records it.

> Three principles run through the whole design. **The ERP is never written to**: the tool reads an extract and answers questions. **Decisions are deterministic**: every finding comes from a fixed rule with a clause, never from an AI opinion. **A human closes every check**: the tool proposes, a reviewer decides.

2. Words you need

Word	Meaning
ERP / PLM	The company's central systems. They hold the drawing register, bills of materials, work orders and purchase orders.
Drawing register	The official list: each drawing number, its current revision and its status.
Title block	The box in the corner of a drawing with its number, revision, status, material and date.
Revision	A version of a drawing (A, B, C ...). Only one is current; older ones are *superseded*.
Extract	A read-only copy of ERP data, exported as files. The tool reads the copy and never writes to the ERP.
BOM	Bill of materials: the parts, materials and quantities that belong to a drawing revision.
Work order	An instruction to the shop floor to make something, released against a drawing revision.
Rule	One fixed check, such as “plate thickness must be at least 6 mm”.
Clause	The numbered paragraph in a standard that a rule comes from. Every finding cites one.
Finding	One problem found on one drawing, with a severity, a clause and the evidence.
Verdict	The tool's proposal for the drawing: Pass, Needs review or Blocked.
Vector PDF / scan	A vector PDF comes from CAD and contains real text. A scan is a photograph of paper and contains only pixels.
OCR	Optical character recognition: software that turns a picture of text into text.

3. Where it sits beside the ERP

The tool is an add-on beside the ERP, not a change to it. An extract of ERP data goes in; verdicts, reports and a gate answer come out.

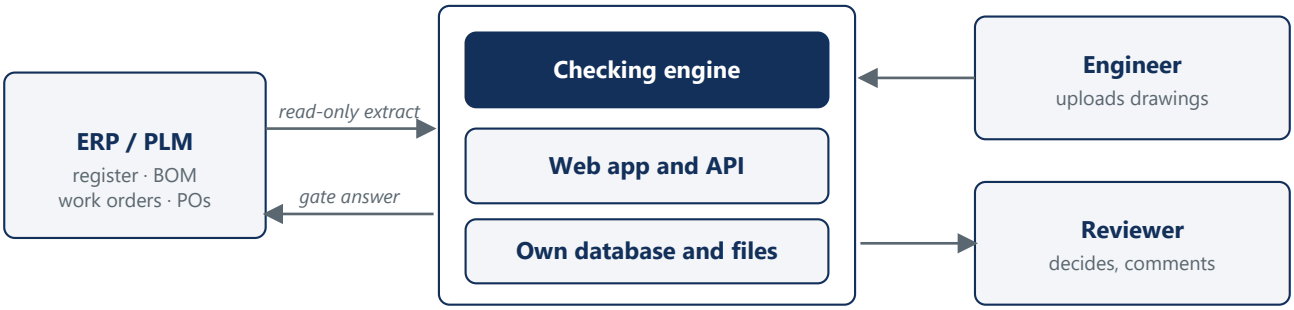


Figure 1. The tool reads an extract from the ERP and answers the gate question. It has its own database for checks, decisions and the audit log.

ERP area	What it holds	Used for
Document / product data management	Drawings, their revisions, engineering change notes	Register check, revision history
Production planning	Work orders, bills of materials	BOM consistency, release gate, change impact
Materials management	Purchase orders	Change impact
Quality / class approvals	What was submitted to class or the customer	Class approval tracking, a warning on the gate

Which ERP GRSE runs, and the exact format of its exports, were not verified. The demo uses seven invented CSV files in `demo/register/` that stand in for the extract.

4. Who uses it

Role	Can do	How the account is created
Engineer	Upload drawings for checking, view everything, comment.	Registers on the sign-in page, then waits for an admin to approve the account.
Reviewer	Upload drawings for checking, view everything, comment, read the audit log. Cannot approve or reject a release, and cannot change rules or the register.	Registers on the sign-in page, then waits for an admin to approve the account.
Admin	Everything. **Only the admin approves or rejects a release.** Also approves and disables accounts, adds or updates drawings in the register, and approves or withdraws rules.	From the logins file only. Nobody can register as admin.

Registration needs approval because a reviewer decides whether steel gets cut. If anyone on the network could create a reviewer account and use it at once, the reviewer's decision would mean nothing.

5. The flow of one check

This is what happens from the moment an engineer chooses a file to the moment a work order is cleared.

1	**Upload** The engineer drops a PDF on the New Check screen. It must be a PDF of at most 25 MB.
2	**Read** The title block, parts list, notes and cited standards are read. A vector PDF is read exactly; a scan is read by OCR.
3	**Confirm (scans only)** For a scan, the values OCR read are shown in editable boxes with low-confidence values marked. The person corrects any mistakes and confirms. The server refuses a scan that was not confirmed.
4	**Look up the register** The drawing number is looked up in the tool's copy of the ERP register, together with its bill of materials.
5	**Run the rules** Each approved rule examines what was read and reports its findings, each with a severity, a clause and the evidence.
6	**Verdict** The worst severity decides: any critical finding means Blocked, otherwise any major means Needs review, otherwise Pass.
7	**Compare with the previous revision** If an earlier revision is available, the changes are listed and the two sheets are overlaid. This is information for the reviewer and does not change the verdict.
8	**BOM and impact** The parts list is set beside the ERP bill of materials, and the work orders and purchase orders affected by the revision change are listed.
9	**Store** The check, its findings, the uploaded file and snapshots of what the ERP said at that moment are saved. The audit log records it.
10	**Reviewer decides** A reviewer opens the report, may comment, and approves or rejects the release. The decision is final and logged.
11	**Release gate** Each work order now shows Clear, Warn or Blocked, with reasons. The ERP (or a person) acts on that answer.

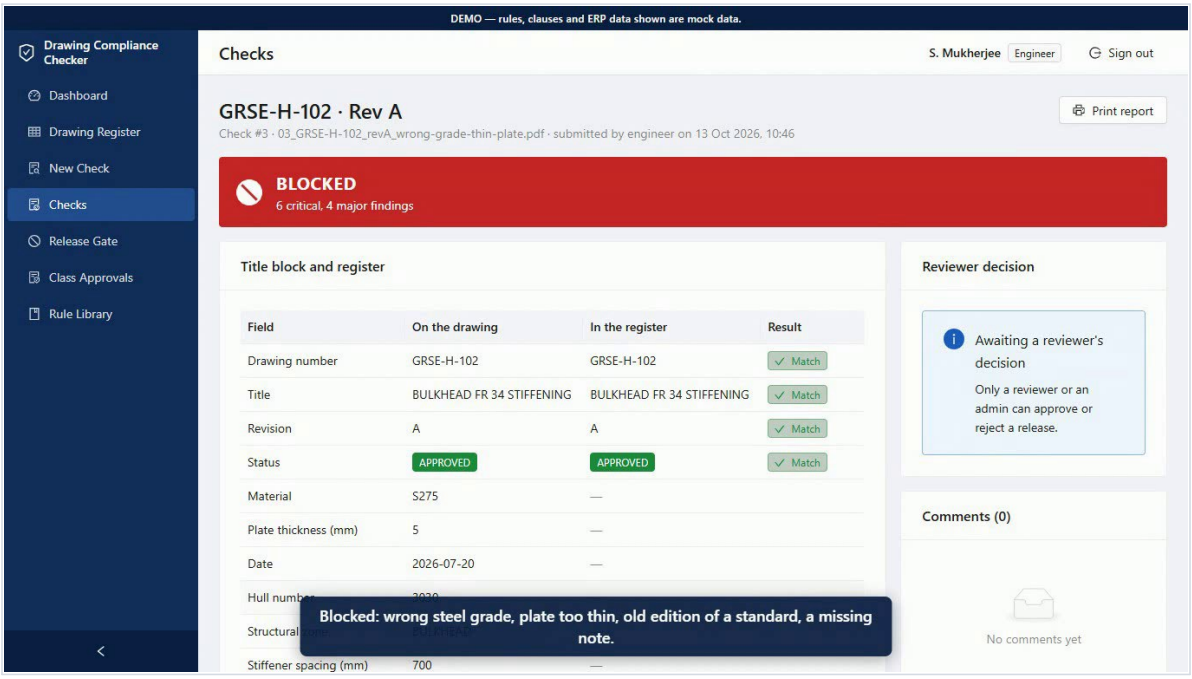


Figure 2. A check report: the verdict, the title block beside the register, and the reviewer's panel.

6. The rules and the verdict

There are twelve **check types** written as code. A **rule** is an entry in a file that picks one check type and gives it parameters, a severity and a clause. Adding a rule of an existing type means adding an entry, not writing code.

A rule runs only while it is **approved**. Approval is given in the tool, on the Rule Library screen, by an admin, and it covers the rule *exactly as written*: if the clause, the severity or a parameter is later edited, the rule stops running until it is approved again. Who approved what, and when, is in the audit log.

Rule	What it catches	Severity	Mock clause
REG-01	A superseded revision, or a drawing that is not in the register	Critical	MOCK-PROC 4.2
REG-02	The title block's status disagrees with the register	Major	MOCK-PROC 4.3
CHK-01	The drawing is not approved for production	Critical	MOCK-CUST 2.1
CHK-02	The drawing cites an old edition of a standard	Major	MOCK-CLS 1.1.4
CHK-03	A material grade that is not on the allowed list	Critical	MOCK-CLS 3.2.1
CHK-04	Plate thinner than the minimum	Critical	MOCK-CLS 3.4.2
CHK-05	A mandatory note is missing	Major	MOCK-CUST 5.3
CHK-06	A recommended note is missing	Minor	MOCK-CUST 5.4
BOM-01	The parts list disagrees with the ERP bill of materials	Major	MOCK-PROC 5.2
CHK-08	Plate thinner than the thickness *calculated* for its structural zone	Critical	MOCK-CLS 4.2.3
CHK-09	Stiffener spacing over the limit for its zone	Major	MOCK-CLS 4.3.1
CHK-10	Thick plate in a zone that is not of a higher steel grade	Critical	MOCK-CLS 3.2.4
CHK-11	A count stated in a note disagrees with the parts list	Major	MOCK-PROC 5.4

Rule	What it catches	Severity	Mock clause
CHK-07	A draft rule, shown only to demonstrate that unapproved rules are skipped	(not run)	MOCK-CUST 5.6

Worst finding	Verdict	Meaning
Critical	**BLOCKED**	The drawing must not be released as it stands.
Major	**NEEDS REVIEW**	A reviewer must look and give a reason either way.
Minor or none	**PASS**	Nothing stops release; the reviewer still confirms.

Rules that depend on where the plate is, and on a calculation

Real class rules do not use one thickness for every plate. They depend on the structural zone (deck, side shell, bulkhead ...), on the ship, and on a formula. The tool handles that shape of rule. Each drawing's title block names its **zone**, its **hull number** and its **stiffener spacing**; the ERP extract gives the ship's rule length for that hull; and a rule can carry one formula per zone.

Zone	Mock formula for required plate thickness (mm)	For the demo ship (L = 142 m, s = 700 mm)
Deck	4.5 + 0.04 × L	10.2 mm
Side shell	5.0 + 0.045 × L	11.4 mm
Bulkhead	4.0 + 0.005 × s	7.5 mm
Machinery seating	8 + 0.05 × L	15.1 mm

So revision A of the deck drawing, with a 10 mm plate, fails; revision B, at 12 mm, passes. The finding shows the working: *"10 mm is below the 10.2 mm required in zone DECK (4.5 + 0.04 * L, with L = 142 m and s = 700 mm)"*. On a longer ship the same drawing would need more. **These formulas and limits are invented.** What is real is that a naval architect can now enter true ones without any change to the code.

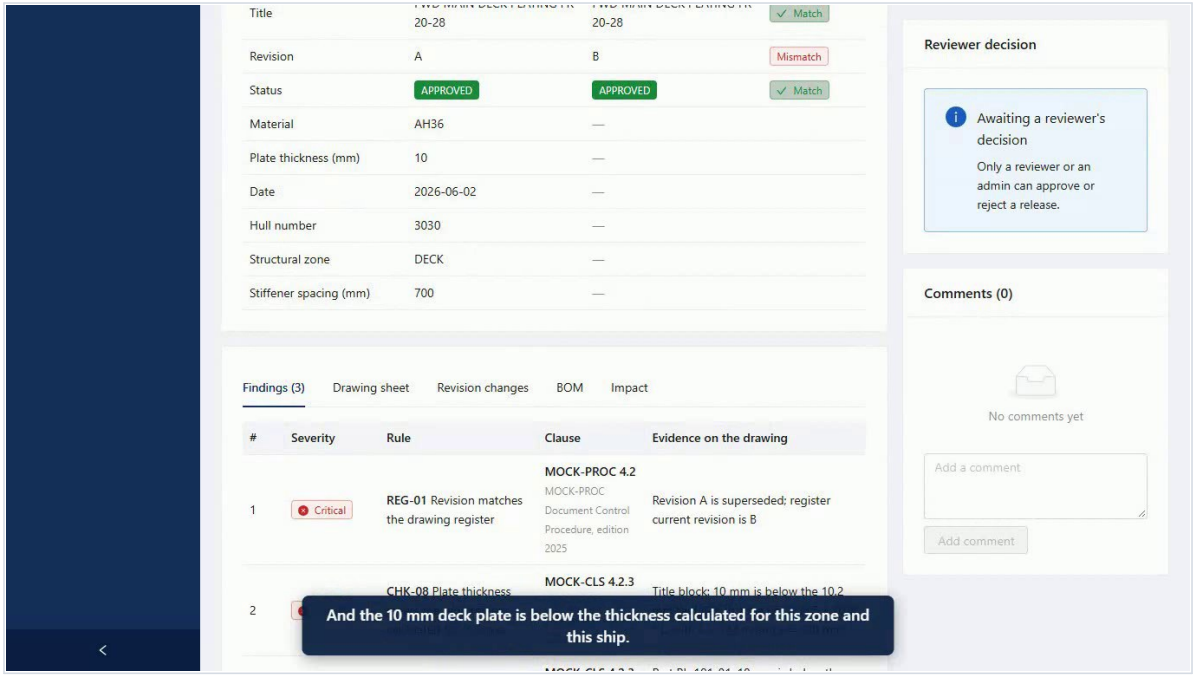


Figure 3a. A calculated finding: the deck plate is thinner than the zone formula requires for this ship.

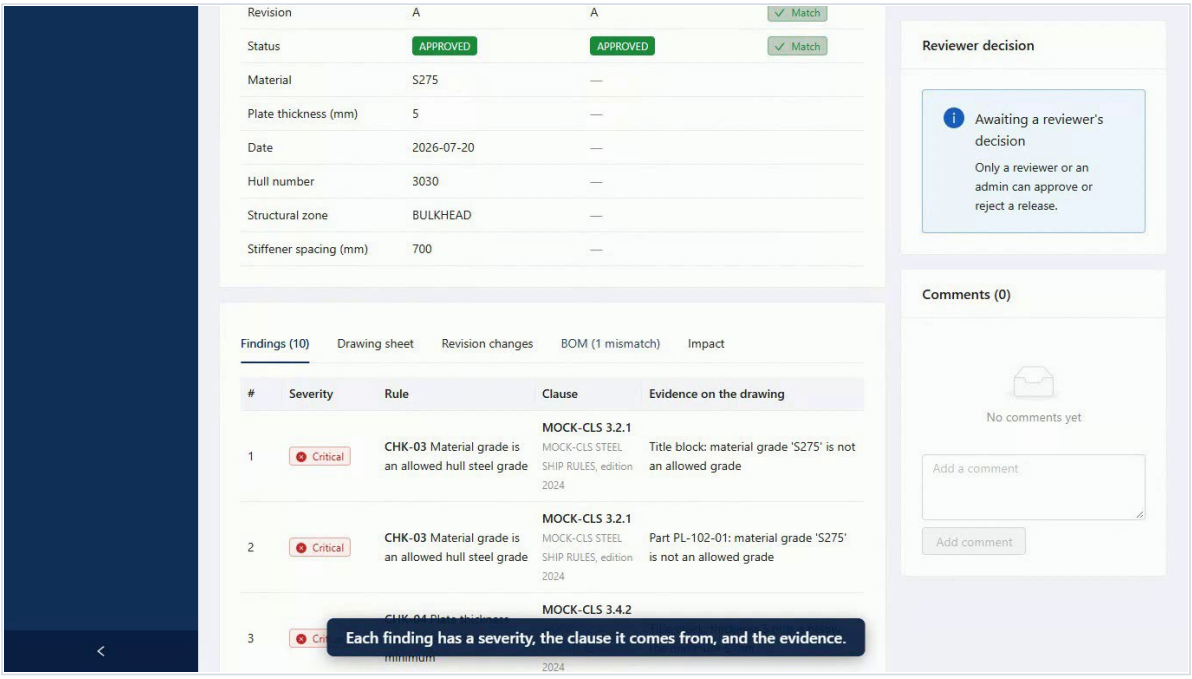


Figure 3. Findings on a non-compliant drawing: severity, rule, clause and the evidence found on the sheet.

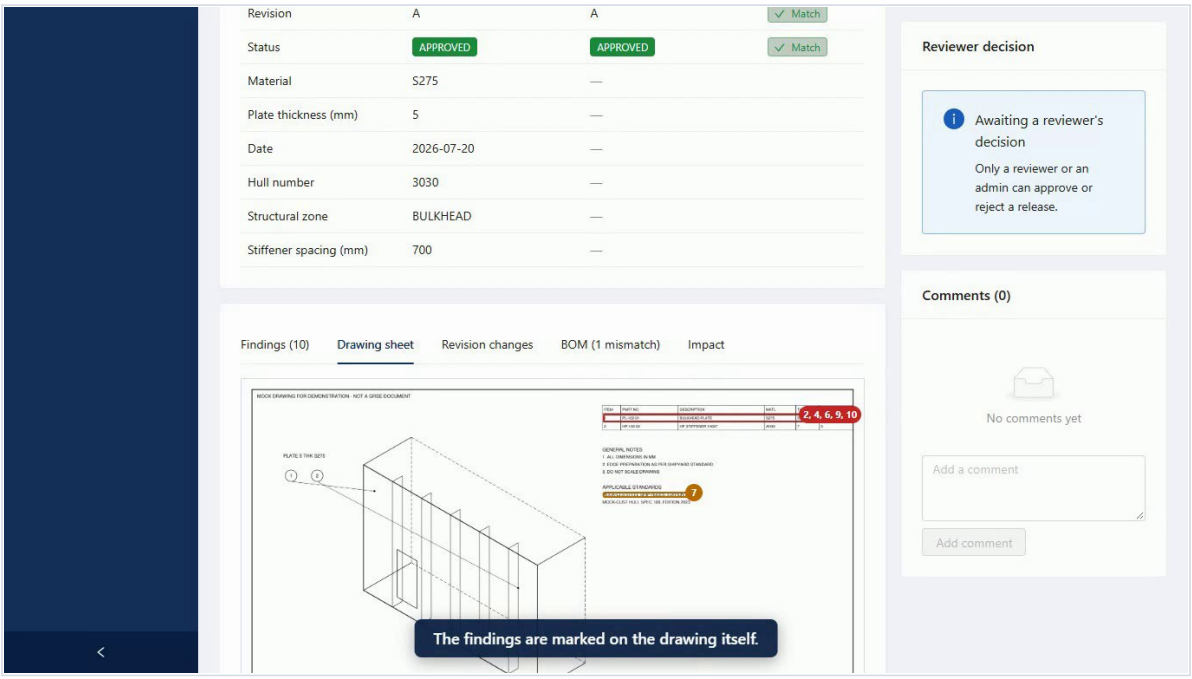


Figure 4. The same findings marked on the drawing sheet. The numbers match the rows of the findings table.

7. Comparing two revisions

"Show exactly what changed" is answered in two ways, because each catches what the other misses.

	The change list	The overlay
Made from	The text: title block field by field, parts list row by row (matched by part number), other text line by line.	The picture: both sheets turned into images of the same size and compared pixel by pixel.
Shows	"PL-101-01 thickness 10 → 12", "BK-101-04 added".	Grey where the sheets agree, red for what is only on the old one, green for what is only on the new one.

	The change list	The overlay
Good at	Saying precisely what a change is.	Catching redrawn lines, which have no text.
Weak at	Geometry: a moved line with no text is invisible to it.	Meaning: something that only moved shows as red and green.

The previous revision is the one uploaded alongside the drawing, or else the latest earlier revision of the same drawing that has already been checked. The ERP's engineering change note for the revision is shown above the comparison, so the reviewer can see whether the drawing changed the way the ERP says it did.

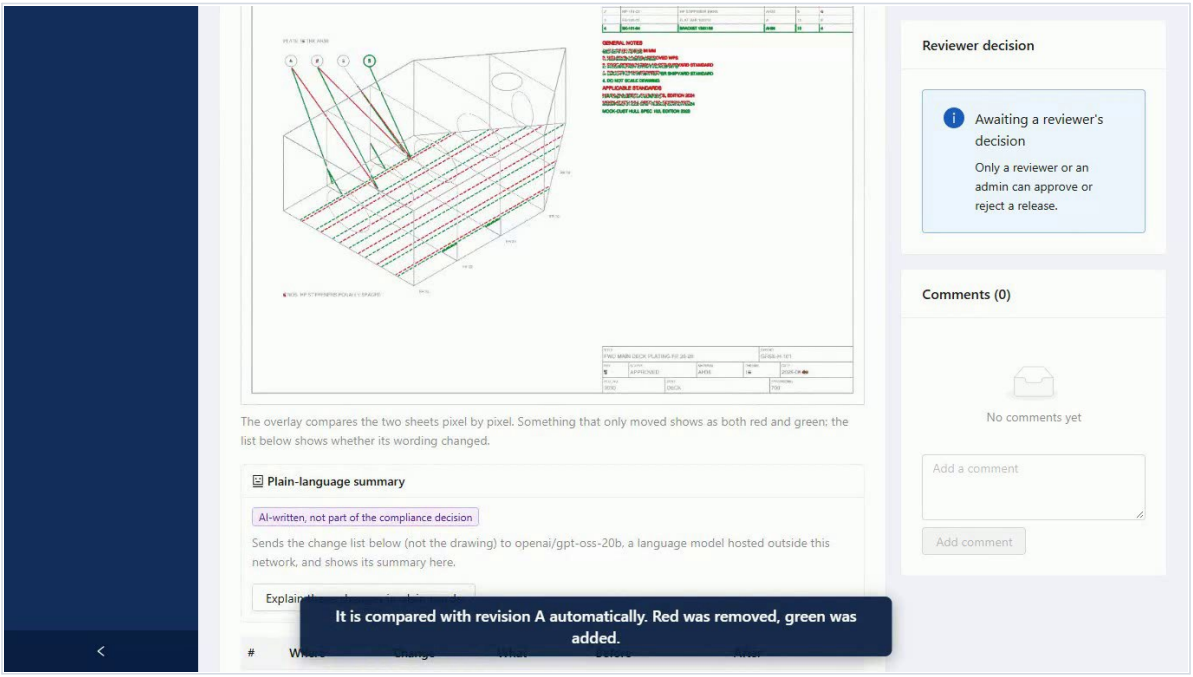


Figure 5. Revision A overlaid on revision B of a bow block: one more stiffener, new brackets and a new parts-list row.

8. Connecting the drawing to the ERP

Does the drawing agree with what will be ordered? (BOM consistency)

The parts list on the drawing is compared with the ERP bill of materials for the same revision, part by part: material, thickness and quantity. A part found on one side only is also reported. Each difference is a finding under rule BOM-01.

Who is affected when a revision changes? (Change impact)

From the parts that changed between two revisions the tool lists the work orders still released against the old revision and the open purchase orders for those parts. For the sample pair it reports: *the change from revision A to B affects 1 work order and 2 open purchase orders*.

May this work order be released? (The release gate)

Fact	Blocked when	Warn when
Register	The revision is superseded, the drawing is unknown, or its status is not approved.	
Check	The latest check of that revision is Blocked and not yet decided.	No check has been run, or it awaits a decision.
Reviewer	The reviewer rejected it.	
Class approval		Not submitted, awaiting, or returned with comments.

A work order is **Clear** only when the revision is current and approved, a reviewer has approved a check of it, and class approval is in place. The gate is worked out afresh every time it is asked, so a decision made a minute ago counts. It is also offered as a web address a system can call, which is how an ERP would ask the question.

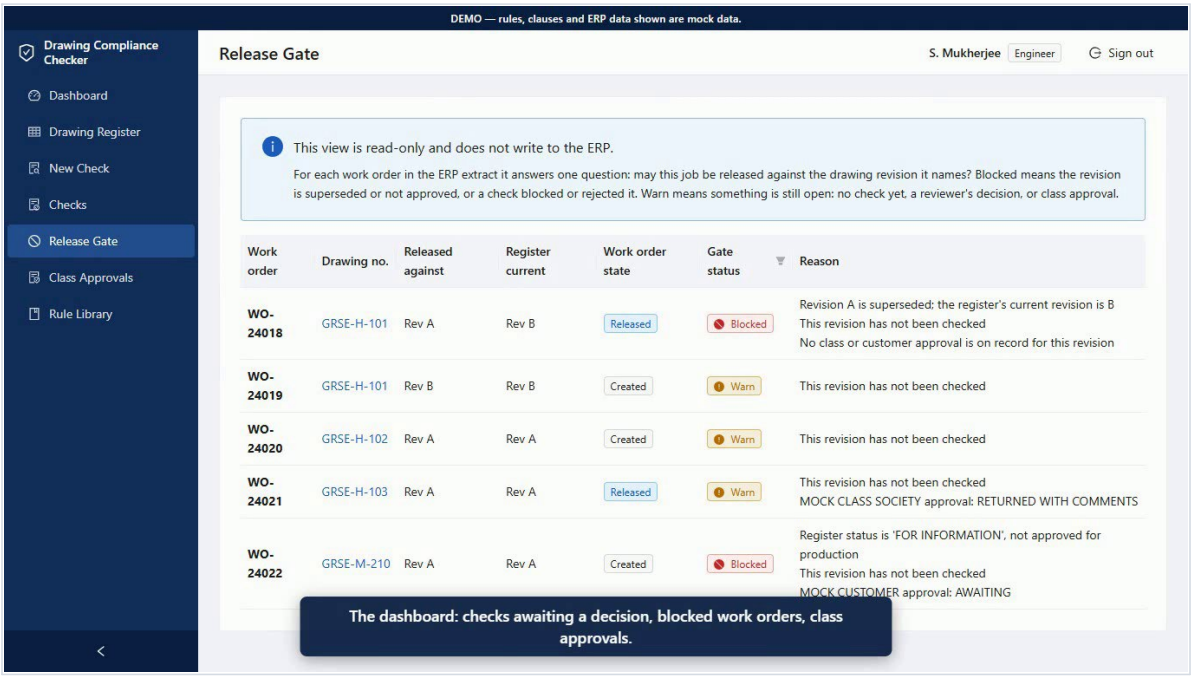


Figure 6. The release gate before any check: two work orders are already blocked by the register alone.

9. Scanned drawings

A scan has no text in it, so it is read by **Tesseract**, an OCR program installed on the same machine; nothing leaves the building. For each word Tesseract gives the text, its position and a confidence from 0 to 100.

- The sheet is read twice with two settings and the results are merged; words below 50% confidence are dropped, because they are nearly always dashed lines mistaken for letters.
- An empty cell in the parts list is read again on its own, enlarged. A whole-page read often skips a lone digit in a table.
- The person uploading must confirm or correct the title-block values before the check runs. Corrections are kept beside what OCR read, and listed in the audit log.
- The parts list and notes are used as read, and the report says so in a banner.
- Before two scans are compared, the older one is shifted and turned until it sits on the newer one.

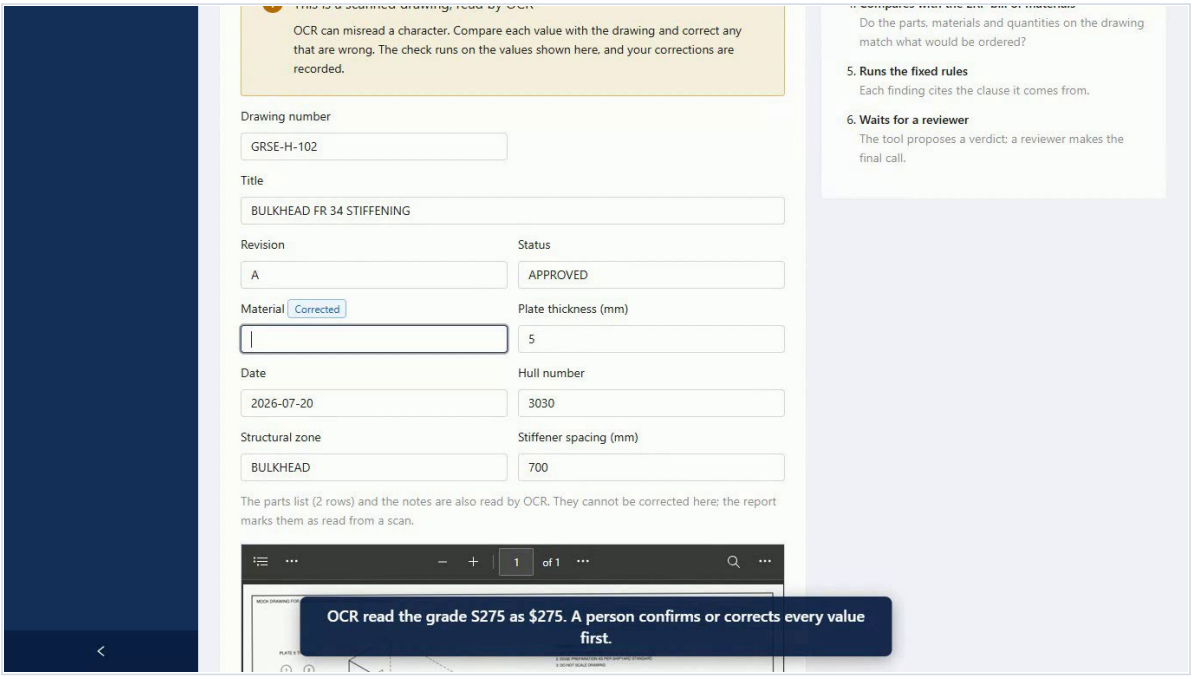


Figure 7. A scanned drawing: OCR read the grade S275 as $275. It is marked low confidence and corrected before the check runs.

10. The optional AI summary

On the Revision changes tab a button can ask a hosted language model for a plain-English summary of the change list. It is a reading aid only. The verdict, the findings, the gate and the decision never look at it.

Risk	What the tool does
The drawing leaves the building	Only the change list is sent: not the PDF and not the drawing number.
It runs without anyone knowing	Nothing is sent until someone presses the button, and the card says where the text will go.
The model invents a number	Every number in the summary is looked up in the change list; any that is not there is called out.
Text on a drawing tries to instruct the model	The model is told the list is data, and its answer is only ever shown as text.
The model gives an opinion on compliance	It is told not to, and the card is labelled “AI-written, not part of the compliance decision”.
No key, or the service is down	The card does not appear, or the button shows the error. Everything else works.

11. The release decision and the audit log

> **Who decides.** In the current version only the **admin** approves or rejects a release. Engineers and reviewers upload drawings, read the reports and comment. Where this document says “the reviewer decides”, read it as the person holding the admin account.

Tool's verdict	Approve for release	Reject
Pass	No reason needed.	Reason required; recorded as an override.

Tool's verdict	Approve for release	Reject
Needs review	Reason required.	Reason required.
Blocked	Reason required; recorded as an override.	No reason needed.

- A check is decided **once**; the decision cannot be edited afterwards.
- A reviewer **cannot decide a check they submitted**. An admin can, and the audit log marks it.
- Anyone signed in can **comment** on a check.
- The **audit log** records sign-ins and failed sign-ins, registrations and approvals, every check, every correction to a scan, every decision, comment, AI summary request and register edit. The tool only ever adds to it.

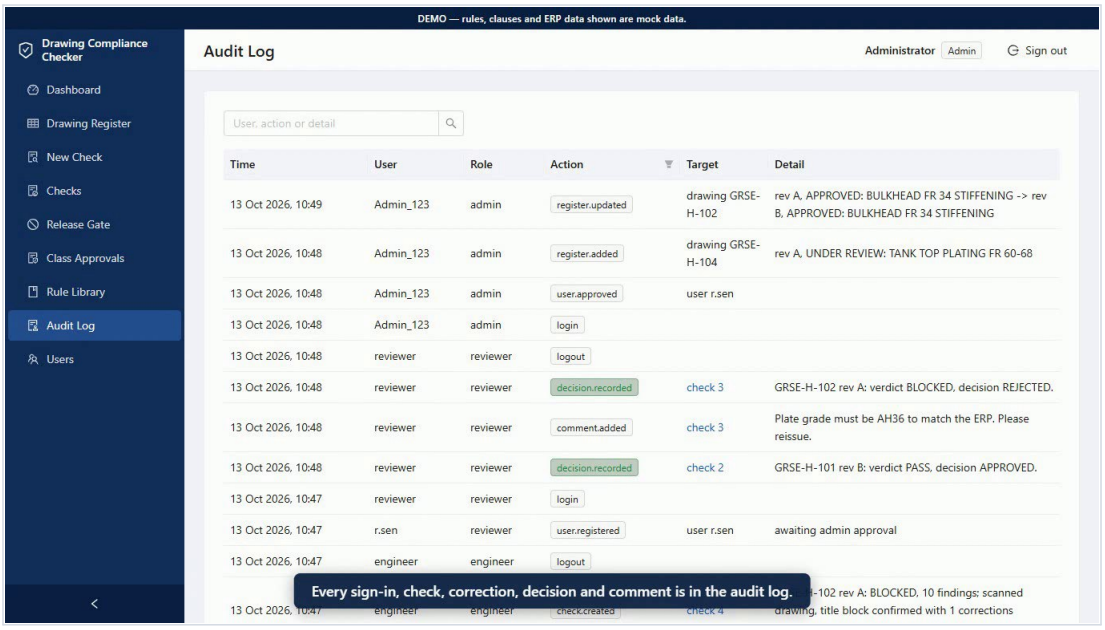


Figure 8. The audit log.

12. Keeping the data safe

- Everything the tool records is in one database file and one uploads folder under `backend/var/`. It survives a page refresh, a restart of the server and a restart of the computer.
- When the code needs different tables, the server upgrades the database at start-up **without losing rows**, and copies the database to a backups folder first.
- An automated test fails if anyone changes a table without writing the matching upgrade step.
- Reloading the ERP extract replaces only what came from the ERP. Checks, decisions and an admin's register edits are left alone.
- The demo's reset option moves old data to an archive folder; it does not delete it.

13. The screens

Screen	What it shows
Sign in	Username and password, and a tab to register as an engineer or a reviewer.
Dashboard	Checks awaiting a decision, blocked work orders, drawings awaiting class approval, recent checks, and what is waiting for you.
Drawing Register	The register with search and filter, each drawing's class approval and latest check. Admin: Add drawing.
Drawing detail	Revision history with change notes, bill of materials, work orders, class approvals, checks. Admin: Update drawing.
New Check	Upload, see what was read (confirm it for a scan), run the check.
Check report	Verdict; title block beside register; tabs for Findings, Drawing sheet, Revision changes, BOM and Impact; decision; comments; print.
Checks	All checks, filterable by verdict and decision.
Release Gate	Every work order: Clear, Warn or Blocked, with reasons.
Class Approvals	What is waiting for class or the customer, and for how long.
Rule Library	The rules with clause, severity, parameters and who approved each. Admin: approve or withdraw a rule.
Audit Log	Reviewer and admin.
Users	Admin: approve, disable and re-enable accounts.

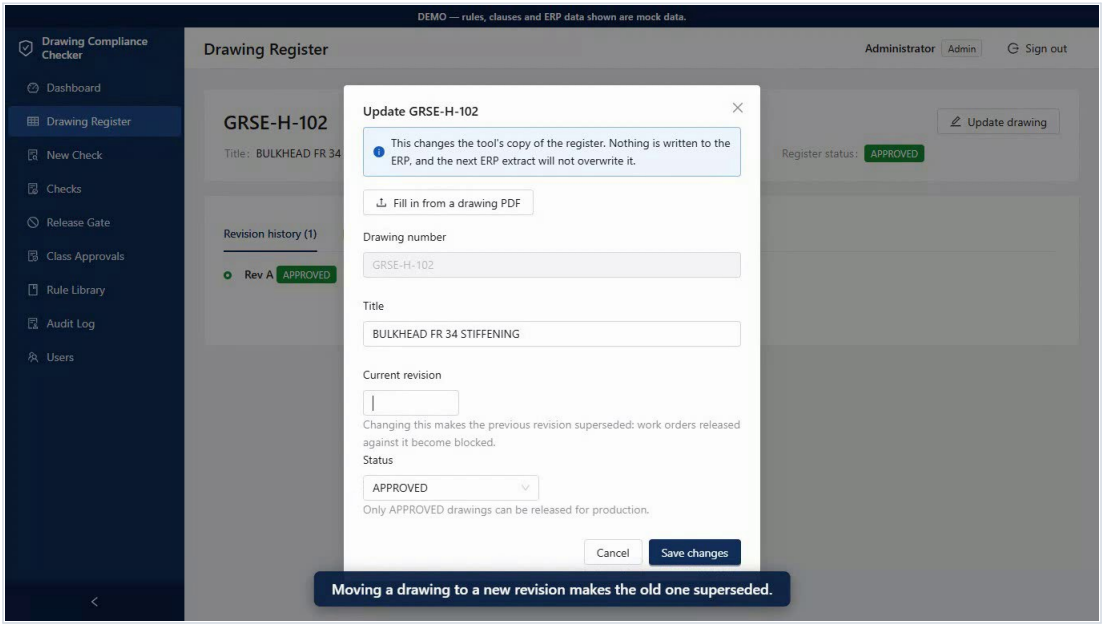


Figure 9. An admin moving a drawing to a new revision. The old revision becomes superseded and its work orders are blocked.

14. The demo kit

Everything needed to try the tool is in the `demo/` folder: a start script, the logins, the mock ERP files, eight mock drawings, a 4.5-minute captioned video and the script that records it.

Drawing	What it shows	Verdict
01 GRSE-H-101 rev A	A superseded revision: the register has moved on to B.	Blocked
02 GRSE-H-101 rev B	A clean drawing, compared automatically with revision A.	Pass
03 GRSE-H-102 rev A	Wrong grade, plate too thin, old edition, missing note, parts list disagrees with the ERP.	Blocked
04 GRSE-H-103 rev A	Old customer-spec edition, a recommended note missing, a quantity differs from the ERP.	Needs review
05 GRSE-M-210 rev A	Issued for information only.	Blocked
06, 07	Scans of drawings 01 and 02.	Blocked, Pass
08	A scan of drawing 03; OCR misreads the grade, for the person to correct.	Blocked

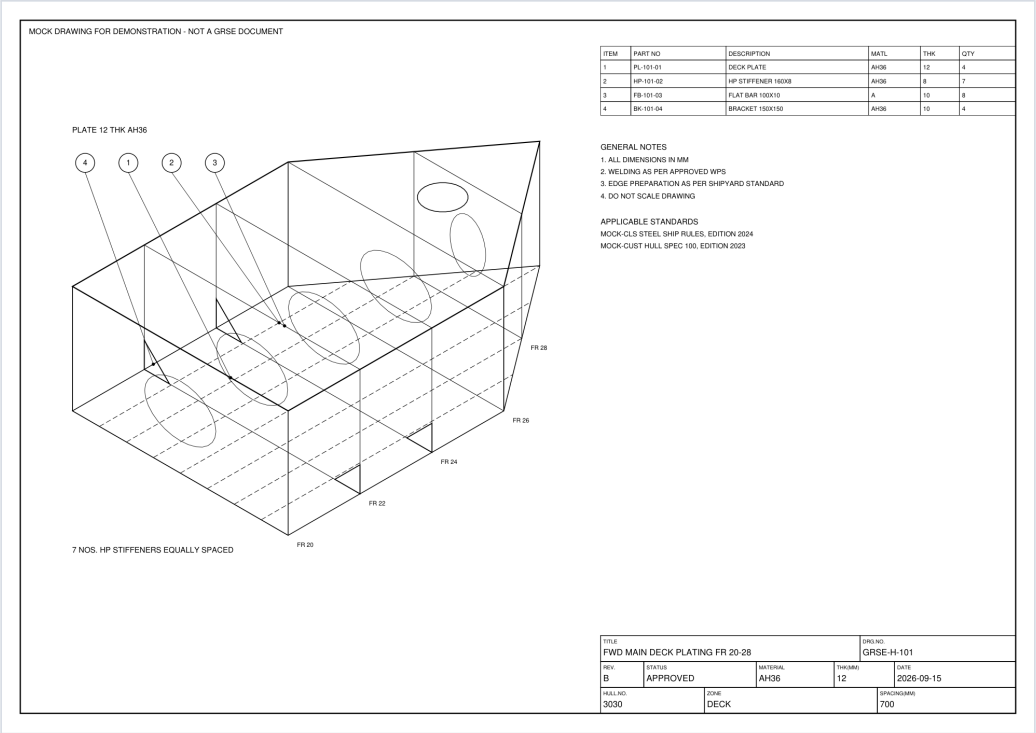


Figure 10. One of the mock drawings: an invented bow block with a title block, parts list, notes and item balloons.

To start it, run `.\demo\run-demo.ps1` from the project folder in PowerShell. A step-by-step walkthrough is in `demo/README.md`.

15. How it was built

Phase	What was added
1	The checking engine: reading the title block and parts list, the register check, the rules, the verdict, and an API.
2	The web app, sign-in with roles, the reviewer's decision and the audit log.
3	Revision comparison: the change list and the red/green overlay.
4	BOM consistency, change impact, the release gate, class approval tracking, the drawing detail page; the demo folder.
—	A durable database with upgrades and backups; the admin role, reviewer registration and comments.
5	Scanned drawings: OCR, confirmation, straightening two scans before comparing.
6	The optional AI summary and its safeguards.
—	The sample video, ship-style mock drawings, and register editing for the admin.
—	Rules that depend on zone and on a calculated formula; rule approval in the tool.

16. What is not done: honest limits

- **The title-block reading is tied to the mock sheet layout.** A real GRSE drawing will not be read correctly until its layout is mapped.
- **The rules are mock.** The clause numbers, limits and formulas are invented. The tool can now apply rules by zone and by formula, and records who approved each, but real content must be written and approved by a naval architect. The demo approvals were made by a setup script, not by an expert.
- **The zone comes from the title block**, one zone per drawing. A drawing covering several zones, or a rule needing more ship data than length, needs more work.
- **The tool reads text and compares pictures.** It does not understand a hull form, a weld symbol or a dimension line.
- **Only the first sheet** of a drawing is read, and revisions are ordered as single letters.
- **The scans tested are clean simulated ones.** Real scans with stamps, handwriting and folds will need more corrections.
- **Sign-in is basic:** no lockout after wrong passwords, and the demo passwords sit in a plain file.
- **The web app has no automated tests**; the backend has 60.
- **Nothing was validated with GRSE data.** That is the next stage.